

# A cortical-proxy observable for emotionally manipulative media: an instrument, a 400-item corpus, and the field bound it yields

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## Abstract

Mean-field opinion models treat media as an external field on a population of coupled states, and that field is almost never measured. A companion paper states what any candidate observable must satisfy for the mechanism to operate:  $\alpha \geq h_c(\beta J)/\Delta X$ . This paper supplies the missing half, an observable computed from content and the spread it attains on a real corpus. Text is spoken, transcribed, embedded and passed to a released trimodal encoder that predicts cortical response; two networks are averaged from that prediction, one associated with affective salience and one with deliberative control, and the observable is their signed difference. The ratio form is undefined for 17.25% of items and is replaced by that difference. On 400 items across four categories, matched on length, the categories separate at  $\eta^2 = 0.1068$ ,  $F = 15.779$ ,  $p = 1.035 \times 10^{-9}$ , against a design powered to detect  $\eta^2 = 0.0268$ . The separation is carried by fear-activating content,  $d = 0.9389$  against neutral, while high-outrage content does not separate,  $d = 0.0300$ . Each category is drawn from one source dataset, so the separation is between corpora and cannot be attributed to framing; we state this before the result rather than in limitations. A second scanning session gives test-retest ICC = 0.8757 and replicates the separation at  $\eta^2 = 0.0839$ , while 13.1% of individual items reverse direction, so the instrument supports category-level statements and no per-item verdict. Calibrating the coupling does not identify it: the fit performs worse than the sample mean out of sample and scales with an unmeasured  $\beta J$ , so no value is quoted. What the measurement does yield is

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the spread,  $\Delta X = 0.1241$ , which converts the companion paper’s inequality into a concrete requirement,  $\alpha \geq 4.2938$  at  $\beta J = 2$ , on any study proposing this mechanism.

*Keywords:* sociophysics, opinion dynamics, external field, media measurement, encoding models, reproducibility

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## 1. Introduction

Statistical-physics treatments of opinion formation routinely include a term representing media. In the mean-field Ising formulation and its relatives, that term is an external field  $h$  acting on every agent, and the modelling result is that beyond a critical magnitude  $h_c(\beta J)$  the population is driven from one consensus to another. The literature on this mechanism is large and the qualitative conclusion is robust within the models.

The field itself is almost never measured. It is assigned, swept, or fitted to an outcome that the model was built to reproduce. A companion paper makes the consequence precise: writing  $h = \alpha X$  for a content observable  $X$  with attainable range  $\Delta X$ , no media-driven transition is possible unless

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}, \quad (1)$$

and this can be evaluated before any data are collected. What it cannot supply is  $\Delta X$ , which is an empirical property of a real observable on real content.

This paper supplies one. We build an instrument that computes a candidate observable from content, apply it to a corpus assembled for the purpose, and report the spread it attains together with what the measurement does and does not support. Three commitments run through the work and are stated here so the reader can hold us to them: the physics stays in front and the encoder is treated as a thermometer rather than a subject; every number is written by a named script into an artifact and read from there rather than transcribed; and the design’s confound is stated before each result it touches, not deferred.

## 2. The observable

### 2.1. Definition

Let  $A_{\text{aff}}$  be the mean predicted activation over a union of regions associated with affective salience, and  $A_{\text{del}}$  the corresponding mean over regions

associated with deliberative control. The proposal for this work specified a ratio,

$$\text{NAA}_{\text{ratio}} = \frac{A_{\text{aff}}}{A_{\text{del}} + \delta}, \quad (2)$$

which is a magnitude asymmetry only when both means sit at or above baseline. The encoder’s outputs are standardised and unconstrained in sign, and a negative denominator makes the ratio change sign for reasons that have nothing to do with the content. Across the corpus reported below the ratio is undefined for 69 of 400 items, 17.25%. This is not an edge case to be dropped, so the analysis is carried by the signed difference

$$X \equiv \text{NAA}_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}, \quad (3)$$

which is defined everywhere. Positive means affective activation is predicted to lead, negative means deliberative activation leads. The reformulation changes what the index means: it reports which system leads, not by what proportion.

## 2.2. What the observable is not

The observable is a *cortical proxy*. The released checkpoint predicts activity on the cortical surface only, so no subcortical structure is measured and none is named. No participant was scanned at any point in this work: every value is a prediction produced by an encoder from content, and the phrase “predicted activation” is used throughout in place of any claim about measured brain activity.

## 3. Instrument and corpus

### 3.1. The cascade

Each item is converted to speech, transcribed with word-level timings, embedded as text and as audio, and passed to **TRIBE v2**, a released tri-modal encoder that predicts fMRI response over 20,484 fsaverage5 vertices [1]. The checkpoint is loaded in the average-subject configuration its own loader forces. Regions of interest are resolved once into vertex index sets and cached, so every item is reduced over identical regions: 1030 vertices for the affective union and 851 for the deliberative union.

### 3.2. Corpus design

The corpus is 400 items in four categories of 100, matched on length. Mean word counts are 167.4, 163.5, 163.7 and 162.2 for fear-activating, high-outrage, neutral and reward-hook items respectively, with standard deviations between 10.2 and 12.8, so length is not a confound.

| Category              | Source              | n       |
|-----------------------|---------------------|---------|
| Fear activating       | ISOT-fake           | 100     |
| High outrage          | SemEval-2019 Task 4 | 100     |
| Reward hook           | Webis-Clickbait-17  | 100     |
| Neutral informational | PubMed, ISOT-true   | 50 + 50 |

Table 1: Corpus composition. Each category is drawn from exactly one source dataset, which is the study’s dominant limitation and is stated before every result it affects.

**The corpus is not source-balanced, and this is the central caveat of the paper.** Category and provenance coincide by construction, so any separation the instrument reports is a separation between corpora and cannot be attributed to framing. Section 4 states this alongside the result rather than after it, and Section 6 identifies which conclusions survive it.

Items were scanned interleaved rather than one category at a time. This turns out to matter, for the reason given in Section 4.3.

## 4. Results

All values below are read from `data/final/rq_answers.json`, written by `scripts/analyze_corpus.py` from the scan output, which was itself verified row-for-row against the source corpus on text, category and identifier.

### 4.1. Separation across categories

The four categories differ at

$$F = 15.779, \quad p = 1.035 \times 10^{-9}, \quad \eta^2 = 0.1068, \quad (4)$$

on  $n = 400$ . The design was fixed before the scan and can detect  $\eta^2 = 0.0268$  at  $\alpha = 0.05$  and power 0.80, so the observed effect is roughly four times the smallest the study could have found.

*Read this with Table 1 in view.* The categories are corpora. The measurement establishes that the instrument assigns systematically different values

to four groups of documents; it does not establish that framing is what it responds to.

| Category vs neutral | t      | p                       | Cohen’s d |
|---------------------|--------|-------------------------|-----------|
| Fear activating     | +6.639 | $3.283 \times 10^{-10}$ | +0.939    |
| Reward hook         | +2.254 | $2.539 \times 10^{-2}$  | +0.319    |
| High outrage        | +0.212 | $8.320 \times 10^{-1}$  | +0.030    |

Table 2: Welch comparisons against neutral informational content. The separation is carried by fear-activating items. High-outrage items do not separate, and that is the category the study’s hypothesis expected to separate most.

Two features of Table 2 deserve emphasis. First, the effect is one category: remove fear-activating items and little remains. Second, high outrage does not separate at all despite being drawn from its own distinct source. If the instrument were simply detecting provenance, every category drawn from a different corpus should move away from the baseline. Three of four moving, in an order consistent with the hypothesis, is a weaker pattern than pure source detection predicts. This is the strongest available argument that the effect is not entirely a corpus artifact, and it is not decisive.

Every category mean is negative, so deliberative activation is predicted to lead throughout. The instrument finds no affective dominance anywhere in this corpus.

#### 4.2. Discrimination against the pre-scan label

Against the **manipulative** label carried by each item before it was scanned, the observable discriminates at  $AUC = 0.6274$ , above the smallest detectable 0.5916. AUC is the headline because it is threshold-free. An  $F_1$  of 0.6680 is obtainable, but its threshold was chosen on the same data it is evaluated on, and it is labelled as fitted in sample wherever it appears.

#### 4.3. Session reliability

A second GPU session rescanned 375 of the 400 items with identical text, identical code and identical region definitions. That pair is a test-retest measurement, and without it every effect above would rest on a single session with no statement of how much of an item’s score is the item and how much is the session.

Three consequences follow, and they cut in different directions.

|                       | $r$    | ICC(2,1) | sd( $\Delta$ ) | sd( $\Delta$ )/sd <sub>between</sub> |
|-----------------------|--------|----------|----------------|--------------------------------------|
| NAA <sub>signed</sub> | 0.8812 | 0.8757   | 0.01006        | 0.475                                |
| A <sub>aff</sub>      | 0.8768 | 0.8696   | 0.01210        | 0.487                                |
| A <sub>del</sub>      | 0.8575 | 0.8544   | 0.01285        | 0.520                                |

Table 3: Agreement between two scanning sessions over the 375 items both cover. ICC is reported beside  $r$  because a session that shifted every score by a constant would still correlate perfectly.

**The separation replicates.** On the second session the four-category ANOVA gives  $\eta^2 = 0.0839$ ,  $F = 11.328$ ,  $p = 3.988 \times 10^{-7}$ , against  $\eta^2 = 0.1054$ ,  $F = 14.576$  for the first session restricted to the same items. Both clear the detectable floor. The result is not an artifact of one session, and the roughly one-fifth attenuation is what additional measurement error does to a ratio of between-group to total variance.

**Per-item direction is not reliable.** Of 375 paired items, 49, or 13.1%, reverse sign between sessions. A statement that a particular item is affective-leading is therefore not supportable at this precision, and no per-item verdict appears anywhere in this paper or in the public interface described in Section 8. Group means over 100 items average this scatter down by an order of magnitude, which is why the category-level result survives.

**Scan order does not carry the separation.** Because the corpus took more than one session, position in the scan is a candidate explanation for the category effect. It is not one. Items were scanned interleaved, so the four categories are balanced across thirds of the scan order,  $\chi^2 = 3.139$ ,  $p = 0.7913$ . The drift is real,  $\eta^2 = 0.0167$  at  $p = 0.0353$ , but it is orthogonal to category by construction: removing scan-third means before the category ANOVA moves the effect only from  $\eta^2 = 0.1068$  to 0.1048,  $p = 1.588 \times 10^{-9}$ . Had the corpus been scanned one category at a time, this drift would have been indistinguishable from the finding.

Neither session is corrected toward the other and the two are not averaged. An average of two sessions would carry a precision that neither run has.

## 5. The physics

### 5.1. Calibration does not identify the coupling

With the observable in hand, the natural step is to estimate  $\alpha$  in  $h = \alpha X$  by fitting the model’s predicted outcome against the corpus. On partial runs

the bootstrap interval for  $\hat{\alpha}$  comfortably included zero. On the completed corpus it no longer does. That is not a result, for two reasons.

The fit performs worse than the sample mean on held-out data, so it has no out-of-sample predictive value: a negative held-out  $R^2$  means the relationship it encodes does not generalise beyond the points used to estimate it. And the estimate scales inversely with  $\beta J$ , which nothing in this study measures, so any single value reported would be a statement about a chosen  $\beta J$  rather than about media.

The coupling is therefore *unidentified*, which is a weaker claim than either a detection or a null. **No value of  $\hat{\alpha}$  is quoted in this paper**, in text, table or caption.

### 5.2. What the spread does yield

The measurement does deliver the quantity Eq. (1) consumes. Across the corpus the observable attains

$$\Delta X = 0.1241, \tag{5}$$

and substituting into the companion paper’s inequality converts a general statement into a concrete requirement.

| $\beta J$ | $h_c(\beta J)$ | $\alpha$ required |
|-----------|----------------|-------------------|
| 1.102     | 0.02101        | 0.1693            |
| 1.496     | 0.20541        | 1.6553            |
| 2.000     | 0.53284        | 4.2938            |

Table 4: The field bound evaluated at three reduced couplings, using the measured spread  $\Delta X = 0.1241$ . Each row states the coupling any proposal of this mechanism would need in order for media to drive a transition through this observable.

This is a necessary condition and not a sufficient one. Clearing it does not establish that media moves populations; falling below it excludes the mechanism within the model.

## 6. What survives the confound

It is worth separating the claims this study can defend from those it cannot, because the two are easy to conflate.

*Not supported.* That the instrument detects manipulation. That framing rather than provenance drives the separation. Any value for the coupling. Any statement about an individual item. Any claim about subcortical structures, or about measured rather than predicted activity.

*Supported.* That the observable attains a measured spread of 0.1241 on real content, which is what the bound consumes and which does not depend on why the spread arises. That the categories separate at an effect four times the detectable floor, and that the separation replicates in an independent session. That high-outrage content does not separate from neutral at this power, which is a clean negative result about the category the hypothesis favoured most. And that the coupling is unidentified on this data, which constrains what a successor may assume.

The confound is removable by design rather than by analysis. Drawing all four categories from a single source, so that provenance is held constant while framing varies, or crossing sources with categories so that the two are statistically separable, would convert the present result into an interpretable one. Neither requires new instrumentation.

## 7. Limitations

1. Category is confounded with source dataset. The dominant limitation, stated in Section 4 as well as here.
2. Single-session scores are noisy:  $ICC = 0.8757$ , with session-to-session standard deviation 47.5% of the between-item standard deviation. Category-level conclusions survive this; per-item ones do not.
3. Predicted activation is not measured activation. Whether the encoder tracks real cortex is unresolved and contested; a published audit reports the released average-subject checkpoint anti-correlated with cortex, and replicating it is unfinished work.
4. Cortical proxy only. The checkpoint cannot speak to subcortical structure.
5. Average-subject configuration, forced by the loader.
6. Out-of-distribution stimuli: the encoder was trained on naturalistic audiovisual material and is given synthesised speech from written text.
7. The corpus is English-language from five specific sources, so the spread is a property of this corpus and the bound inherits that.



## 8. Conclusion

Media enters mean-field opinion models as a field that is almost never measured. This paper measured a candidate: an observable computed from content through a released cortical encoder, applied to 400 items, with its spread, its reliability and its failure modes reported rather than assumed. The instrument separates four corpora at an effect it was powered to find, the separation replicates across sessions, and it is confounded with provenance in a way no analysis of this corpus can undo. The coupling it was built to calibrate is unidentified. What survives is the spread, and through the companion paper’s inequality a requirement,  $\alpha \geq 4.2938$  at  $\beta J = 2$ , that any future proposal of this mechanism must meet.

### Code and data availability

The pipeline, corpus builder, analysis and figure scripts, and the JSON and CSV artifacts every number above is read from, are public at <https://github.com/brn-mwai/monarch>. The instrument runs at <https://monarch.brianmwai.com>, carrying the full corpus with each item’s text, its pre-scan labels and its measured values, and reporting no per-item verdict for the reason given in Section 4.3. Figures showing predicted responses are generated by `scripts/render_brain_figures.py` from the saved per-vertex maps, each stating its threshold, rather than captured from the interface.

### CRediT authorship contribution statement

**Brian Mwai:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. The author is the sole contributor to this work.

### Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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